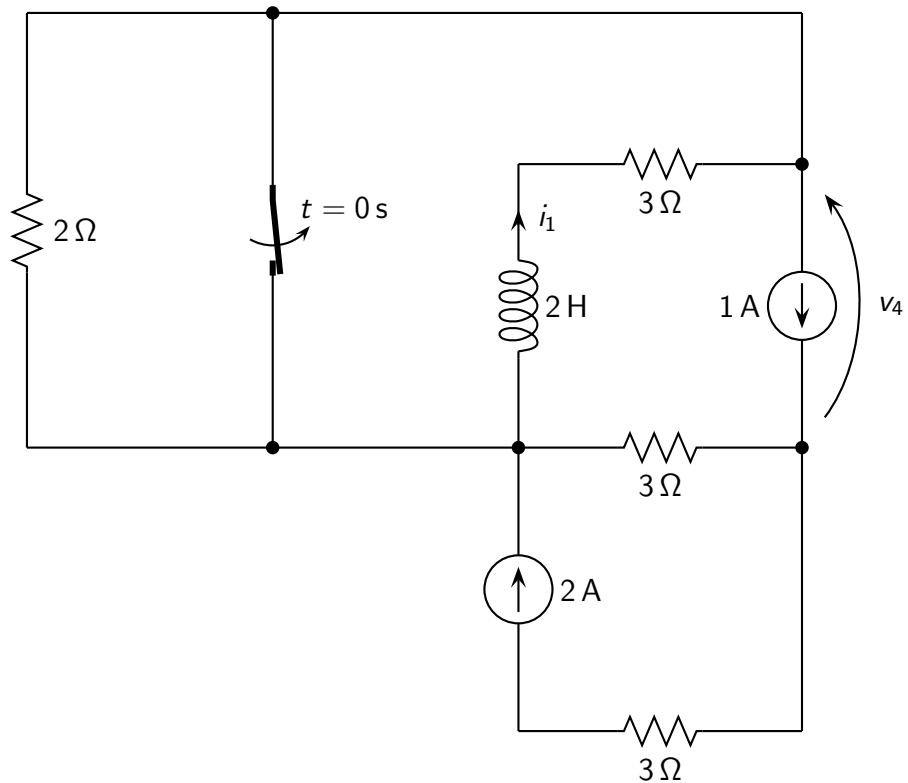


Problem: find the expression of $i_1(t)$, $v_4(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{2}{5}$ s;
- Initial conditions before switching: $i_1(0^-) = 0$ A;
- Initial conditions after switching: $i_1(0^+) = 0$ A; $v_4(0^+) = 1$ V;
- Steady-state solution: $i_1(\infty) = \frac{2}{5}$ A; $v_4(\infty) = \frac{9}{5}$ V;
- Solution for $t > 0$:

$$i_1(t) = \left(\frac{-2}{5} e^{-t/\tau} + \frac{2}{5} \right) \text{ A}$$

$$v_4(t) = \left(\frac{-4}{5} e^{-t/\tau} + \frac{9}{5} \right) \text{ V}$$